

The Chain Rule and Backpropagation

Deep Learning

Daniel E. Acuna

Associate Professor, University of Colorado Boulder

Contents of This Video

In this video, we will cover:

- The challenge of computing gradients in neural networks
- Review of the chain rule from calculus
- Applying the chain rule to a single neuron
- A complete numerical example
- Why it's called “backpropagation”

The Challenge

Loss depends on weights through many intermediate steps:

$$L = L(\hat{y})$$

$$\hat{y} = \sigma(z)$$

$$z = wx + b$$

We need $\frac{\partial L}{\partial w}$, but L doesn't directly depend on w !

Solution: Chain rule traces how changes in w propagate to L

The Chain Rule: Quick Review

If $y = f(g(x))$, then:

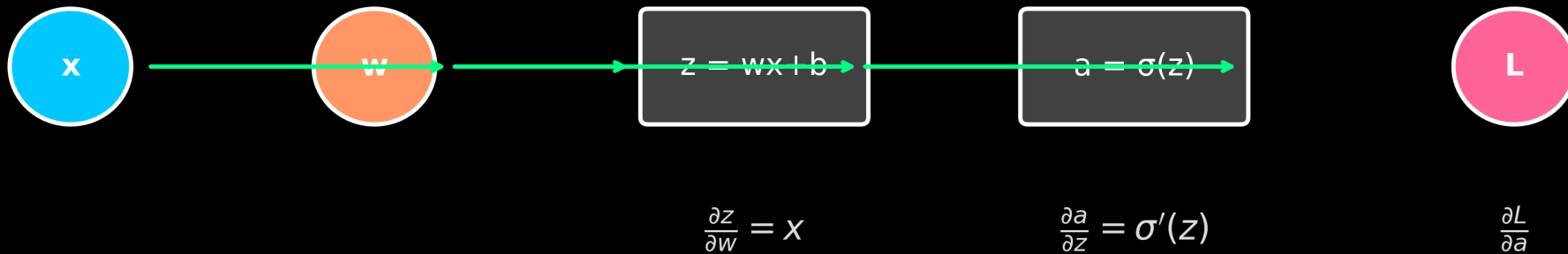
$$\frac{dy}{dx} = \frac{dy}{dg} \cdot \frac{dg}{dx}$$

Example: $y = (2x + 1)^3$

- Let $g = 2x + 1$ and $y = g^3$
- $\frac{dg}{dx} = 2$ and $\frac{dy}{dg} = 3g^2$
- $\frac{dy}{dx} = 3g^2 \cdot 2 = 6(2x + 1)^2$

Applying the Chain Rule to a Neuron

Chain Rule: $\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w}$



Computing Each Piece

For a neuron with sigmoid and binary cross-entropy:

$$z = wx + b, \quad a = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$L = -[y \log(a) + (1 - y) \log(1 - a)]$$

The derivatives:

- $\frac{\partial z}{\partial w} = x$
- $\frac{\partial a}{\partial z} = \sigma(z)(1 - \sigma(z)) = a(1 - a)$
- $\frac{\partial L}{\partial a} = -\frac{y}{a} + \frac{1-y}{1-a}$

Putting It Together

The complete gradient:

$$\begin{aligned}\frac{\partial L}{\partial w} &= \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w} \\ &= \left(-\frac{y}{a} + \frac{1-y}{1-a} \right) \cdot a(1-a) \cdot x\end{aligned}$$

After simplification:

$$\frac{\partial L}{\partial w} = (a - y) \cdot x$$

Beautiful result: Error times input!

Numerical Example

Setup: $x = 2$, $w = 0.5$, $b = 0$, $y = 1$ (true label)

Forward pass:

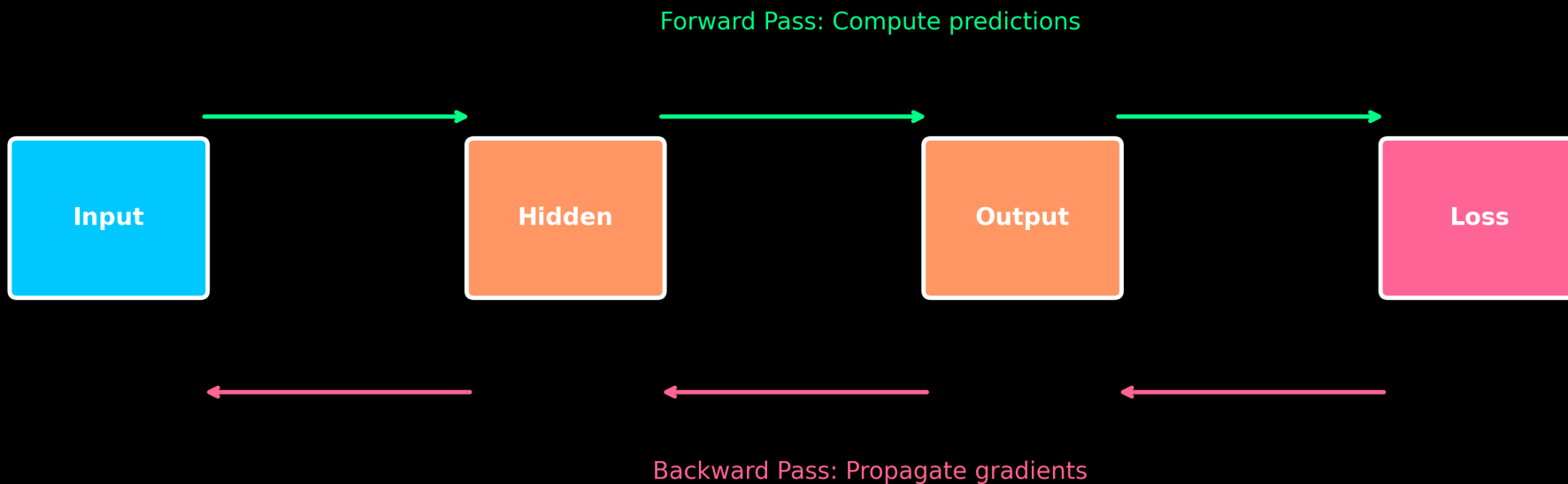
- $z = wx + b = 0.5 \times 2 + 0 = 1$
- $a = \sigma(1) = \frac{1}{1+e^{-1}} \approx 0.731$

Computing gradients:

- $\frac{\partial z}{\partial w} = x = 2$
- $\frac{\partial a}{\partial z} = a(1 - a) = 0.731 \times 0.269 \approx 0.197$
- $\frac{\partial L}{\partial a} = -\frac{1}{0.731} + 0 \approx -1.368$

$$\frac{\partial L}{\partial w} = (-1.368) \times 0.197 \times 2 \approx -0.539$$

Why “Backpropagation”?



Gradients flow backward from loss to inputs, layer by layer

What We've Covered

In this video, we've learned:

- The chain rule lets us compute gradients through composed functions
- $\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial w}$
- Each piece is computed locally at each step
- For sigmoid + cross-entropy: $\frac{\partial L}{\partial w} = (a - y) \cdot x$
- “Backpropagation” = gradients flow backward through the network